

Electrical Circuits I

ENGR 2405

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Chapter 05 – Operational Amplifiers

Austin Community College

Overview

- Introduction
- Math Functions
- Instrumentation Amplifiers

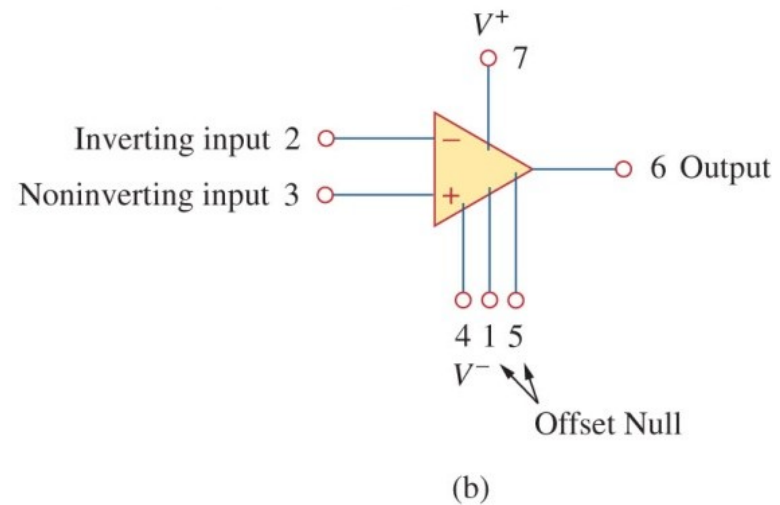
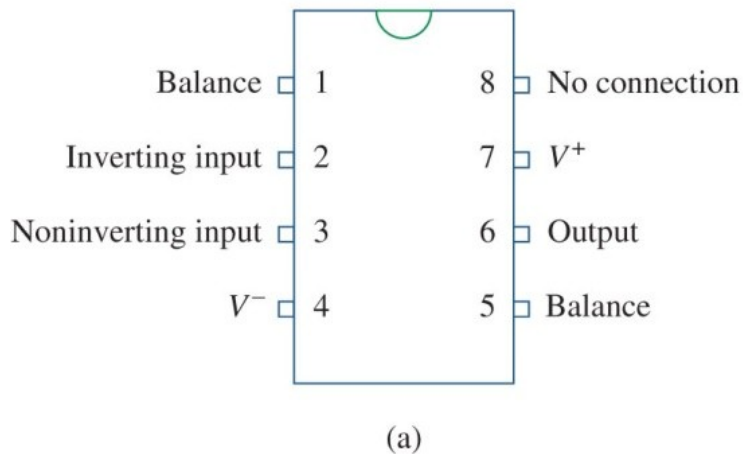
Operational Amplifier (Op Amp)

- Acts like a VCVS
- Can be made into other dependent sources by combining with other elements
- Performs mathematical operations on analog signals

Op Amp Pinout

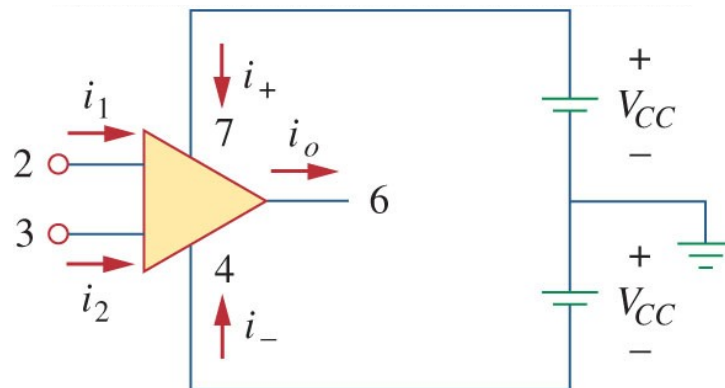
There are five terminals found on all op-amps

- Inverting input
- Non-inverting input
- Output
- Positive and negative power supplies



Powering an Op-amp

- As an active element, the op-amp requires a power source
- Often in circuit diagrams the power supply terminals are not shown
 - It is taken for granted that they must be connected
 - Most op-amps use two voltage sources, with a ground reference between them
 - This gives a positive and negative supply voltage



Output Voltage

- The voltage output of an op-amp is proportional to the difference between the non-inverting and inverting inputs

$$v_o = Av_d = A(v_2 - v_1)$$

- Here, A is called the open loop gain
- Ideally it is infinite
- In real devices, it is still high: 10^5 to 10^8 volts/volt

TABLE 5.1

Typical ranges for op amp parameters.

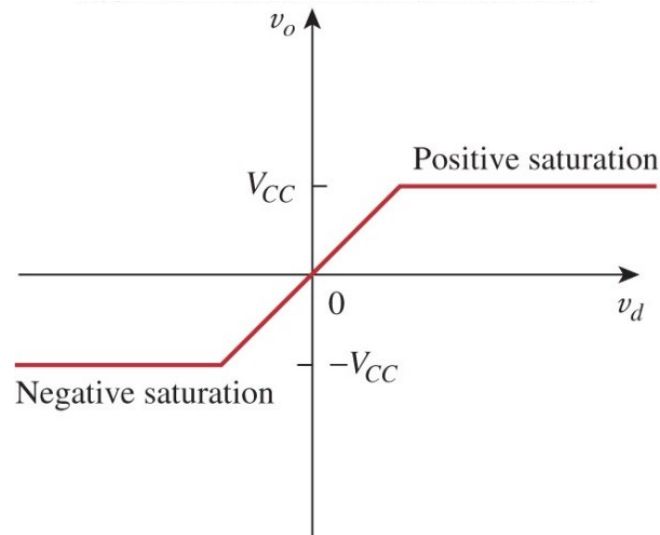
Parameter	Typical range	Ideal values
Open-loop gain, A	10^5 to 10^8	∞
Input resistance, R_i	10^5 to $10^{13} \Omega$	$\infty \Omega$
Output resistance, R_o	10 to 100 Ω	0 Ω
Supply voltage, V_{CC}	5 to 24 V	

Feedback

- Op-amps take on an expanded functional ability with the use of feedback
 - The output of the op-amp is “fed back” into the inverting terminal
 - Depending on what elements this signal passes through the gain and behavior of the op-amp changes
- Feedback to the inverting terminal is called “negative feedback”
- Positive feedback would lead to oscillations

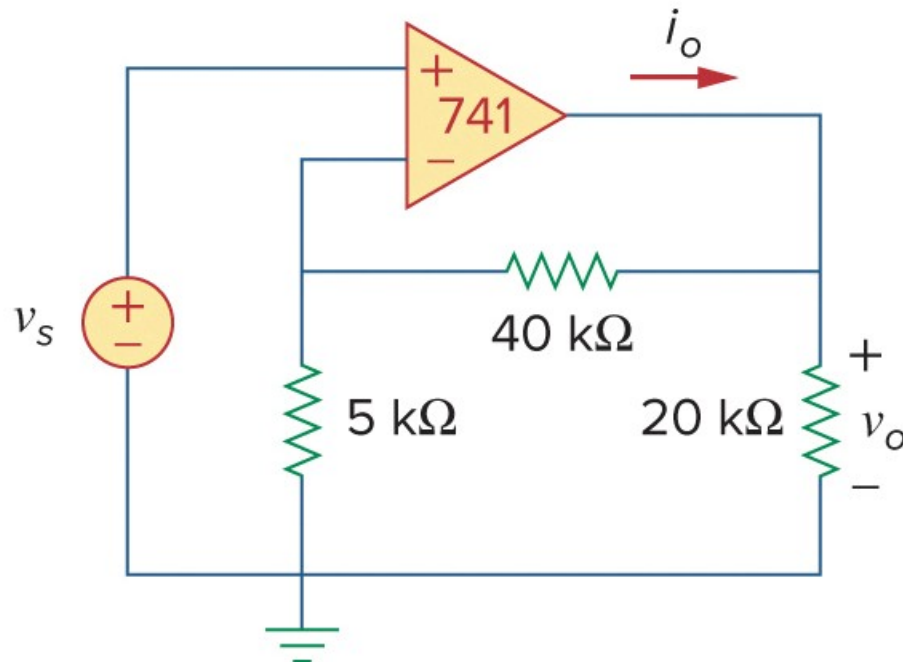
Voltage Saturation

- As an ideal source, the output voltage would be unlimited, but in reality, one cannot expect the output to exceed the supply voltages
- When an output exceeds the voltage range, the output remains at either the maximum or minimum supply voltage, aka saturation
- Outputs between these limiting voltages are referred to as the linear region



Practice Problems

5.1 Calculate the closed-loop gain v_o/v_s and find i_o when $v_s = 1V$



Ideal Op Amp

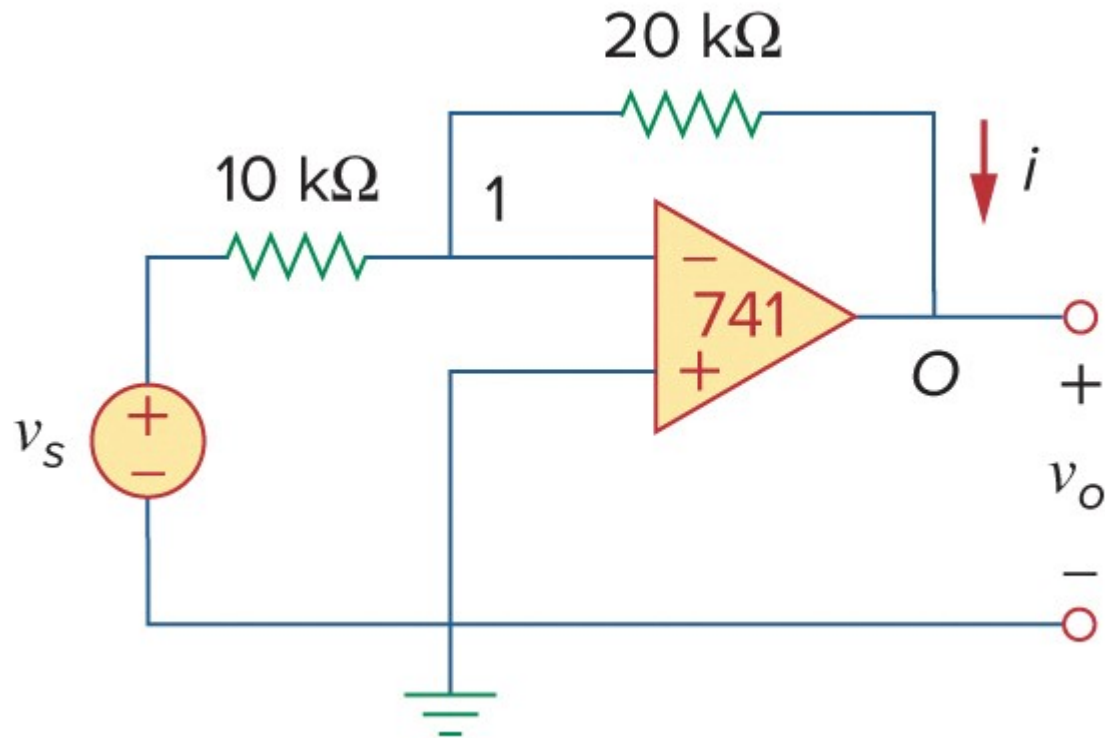
- We give certain attributes to the ideal op-amp
 - infinite open-loop gain
 - Infinite impedance on inputs
 - Zero impedance on output
- From TT, this means it is load independent

Ideal Op Amp

- Many modern op-amp come close to the ideal values:
 - Most have very large gains ($>1\text{M}$)
 - Input impedances are in the $\text{G}\Omega$ to $\text{T}\Omega$ range
 - $\Rightarrow$ No current flows into inputs
- In negative feedback, opamp adjusts output to keep $\Delta V=0$

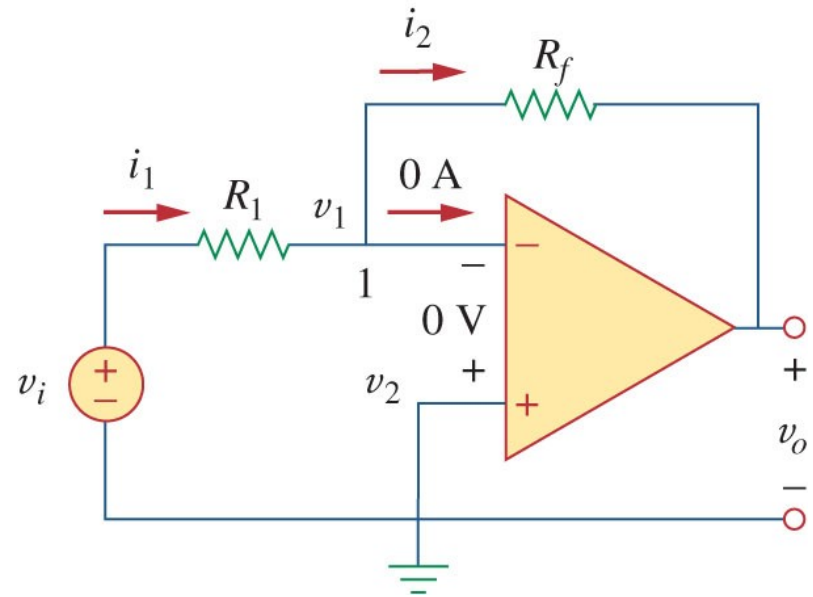
Practice Problems

5.2 Calculate the closed-loop gain v_o/v_s and find i when $v_s = 2V$



Inverting Amplifier

- The first useful op-amp circuit that we will consider is the inverting amplifier
- Here the non-inverting input is grounded
- The inverting terminal is connected to the output via a feedback resistor, R_f
- The input is also connected to the inverting terminal via another resistor, R_1



Inverting Amplifier

- By applying KCL to node 1 of the circuit, one can see that:

$$i_1 = i_2 \Rightarrow \frac{v_i - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$

- Non-inverting terminal is grounded
- Negative feedback $\Rightarrow v_1$ is also zero volts
- This yields:

$$\frac{v_i}{R_1} = -\frac{v_o}{R_f}$$

Inverting Amplifier

- This can be rearranged to show the relationship between the input and output voltages:

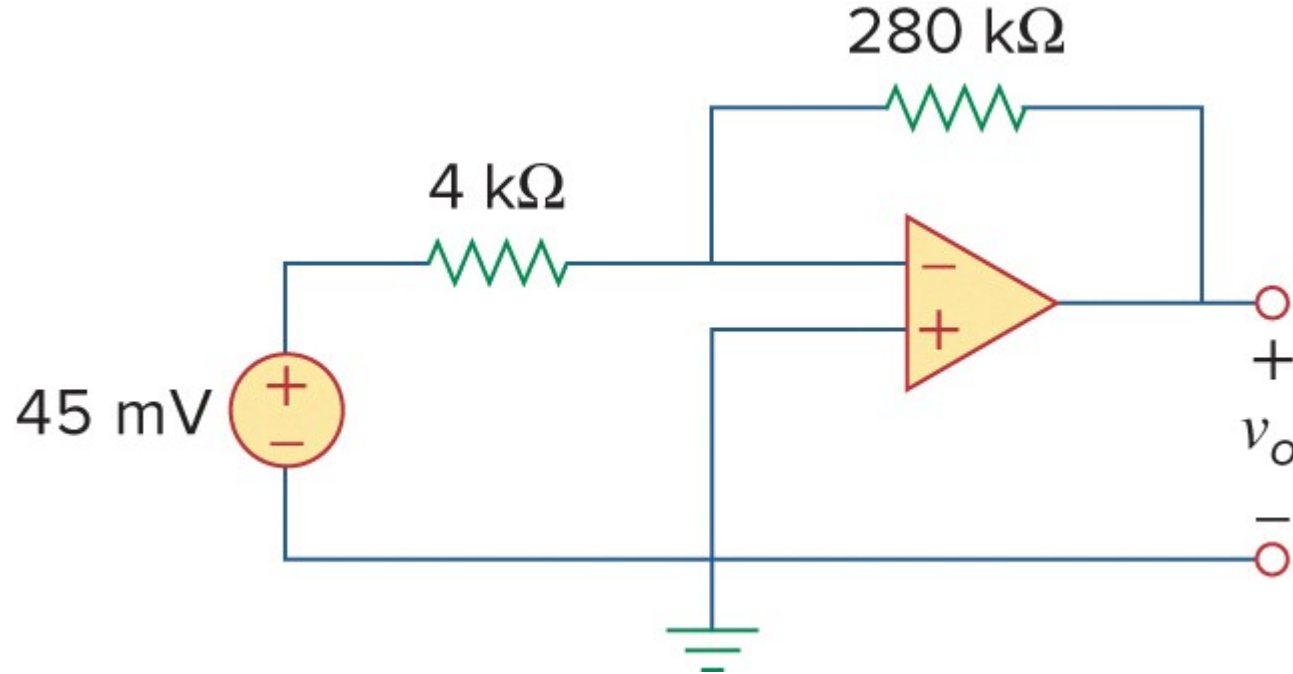
$$v_O = -\frac{R_f}{R_1} v_i$$

From this one can see that:

- The gain is the ratio of the feedback resistor and R_1
- The polarity of the output is the reverse of the input, thus the name “inverting” amplifier

Practice Problems

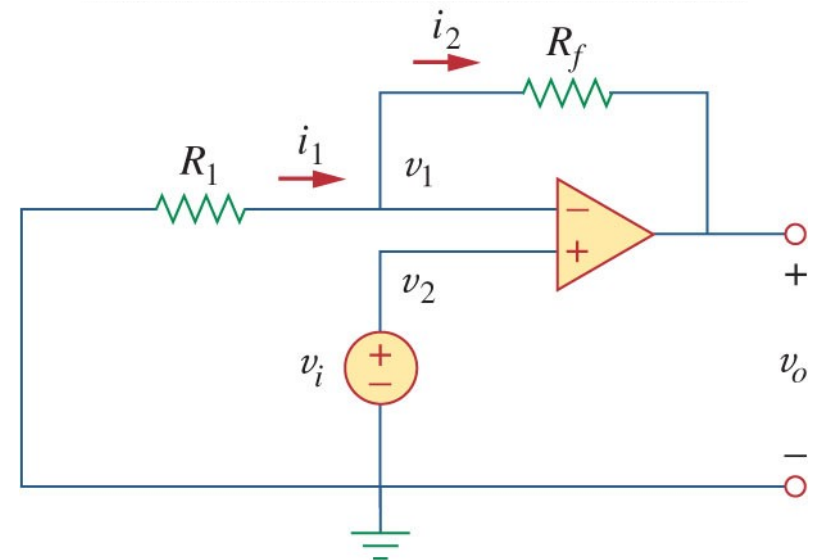
5.3 Calculate v_o and find current through feedback resistor



Non-Inverting Amplifier

- Another important op-amp circuit is the non-inverting amplifier
 - The basic configuration of the amplifier is the same as the inverting amplifier
 - Except that the input and the ground are switched
- Once again applying KCL to the inverting terminal gives:

$$i_1 = i_2 \Rightarrow \frac{0 - v_1}{R_1} = \frac{v_1 - v_o}{R_f}$$



Non-Inverting Amplifier

- There is once again negative feedback in the circuit, thus we know that the input voltage is present at the inverting terminal
- This gives the following relationship:

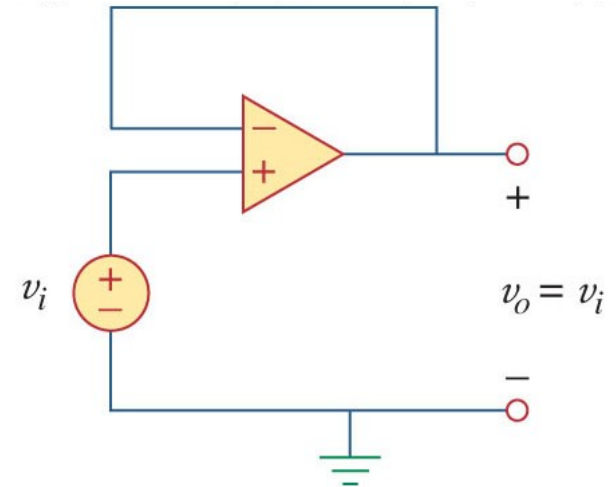
$$-\frac{V_i}{R_1} = \frac{V_i - V_o}{R_f}$$

- The output voltage is thus:

$$V_o = \left(1 + \frac{R_f}{R_1}\right) V_i$$

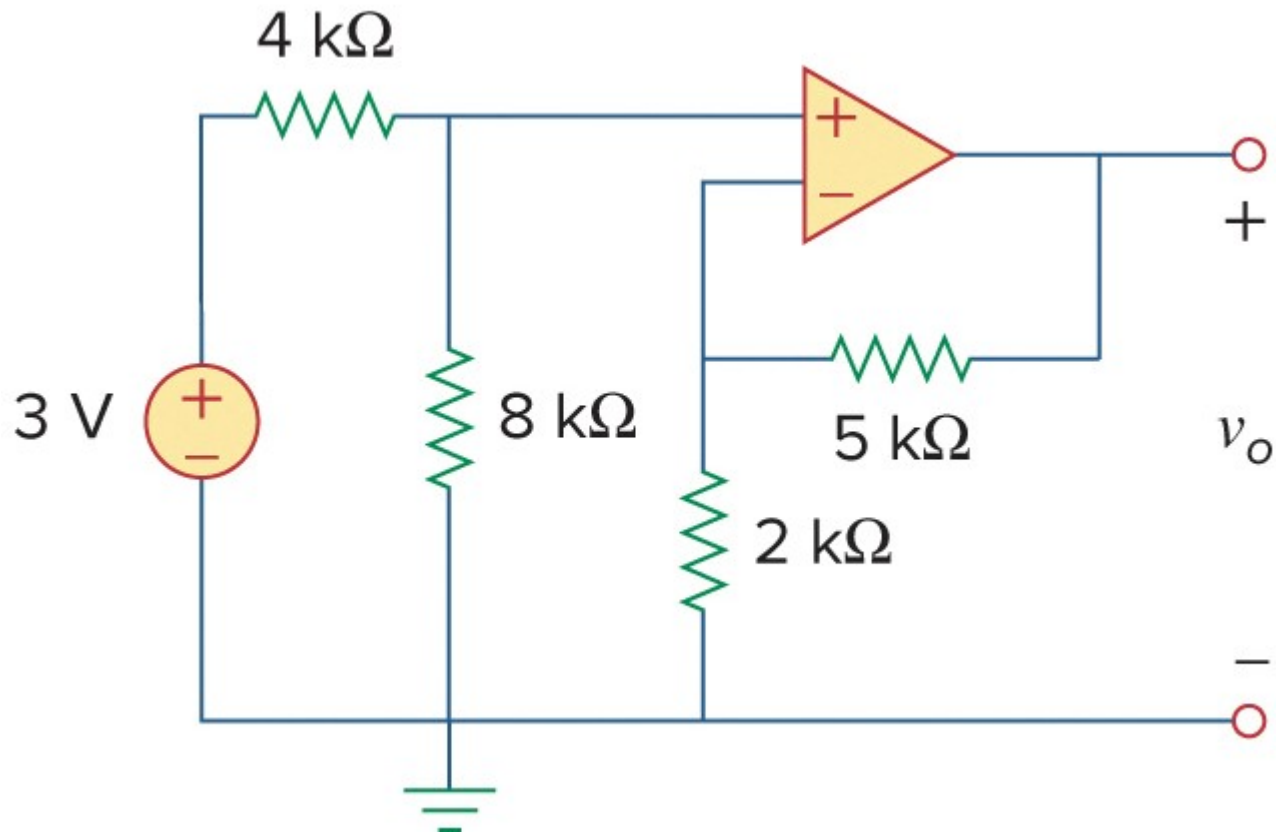
Non-Inverting Amplifier

- Gain is positive => non-inverting
- Note: it can never go below 1
- Replace feedback R with a wire for a gain of 1
- This configuration is called a voltage follower or a unity gain amplifier
- It is good for separating two circuits while allowing a signal to pass through



Practice Problems

5.5 Calculate v_o



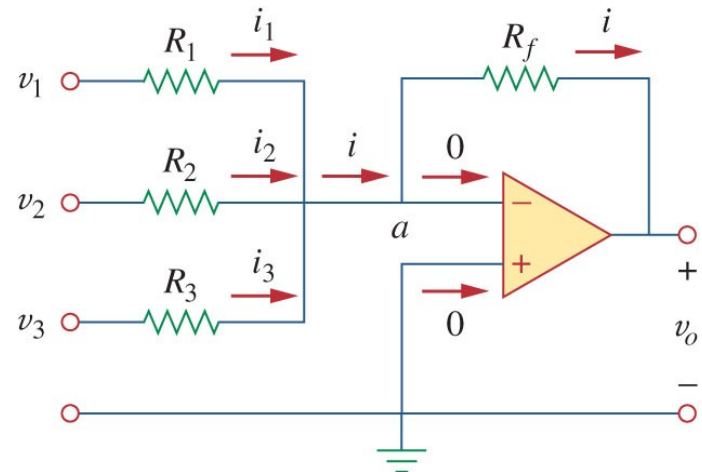
Summing Amplifier

Aside from amplification, the op-amp can be made to do addition very readily

If one takes the inverting amplifier and combines several inputs each via their own resistor:

- The current from each input will be proportional to the applied voltage and the input resistance.

$$i_1 = \frac{(v_1 - v_a)}{R_1} \quad i_2 = \frac{(v_2 - v_a)}{R_2} \quad i_3 = \frac{(v_3 - v_a)}{R_3}$$



Summing Amplifier

- At the inverting terminal, these current will combine to equal the current through the feedback resistor

$$i_a = \frac{(v_a - v_o)}{R_f}$$

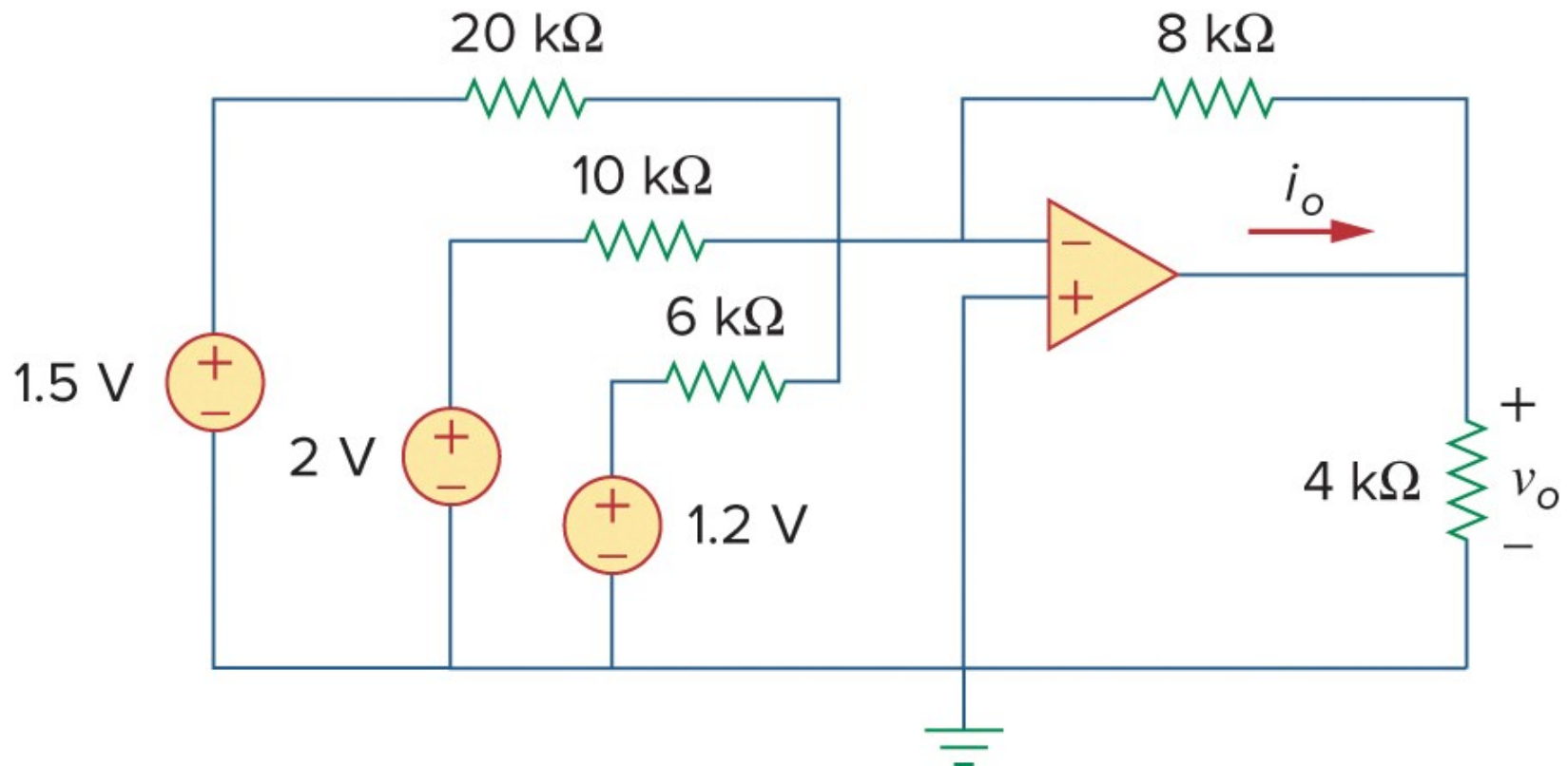
- This results in the following relationship:

$$v_o = - \left(\frac{R_f}{R_1} v_1 + \frac{R_f}{R_2} v_2 + \frac{R_f}{R_3} v_3 \right)$$

- Note that the output is a weighted sum of the inputs

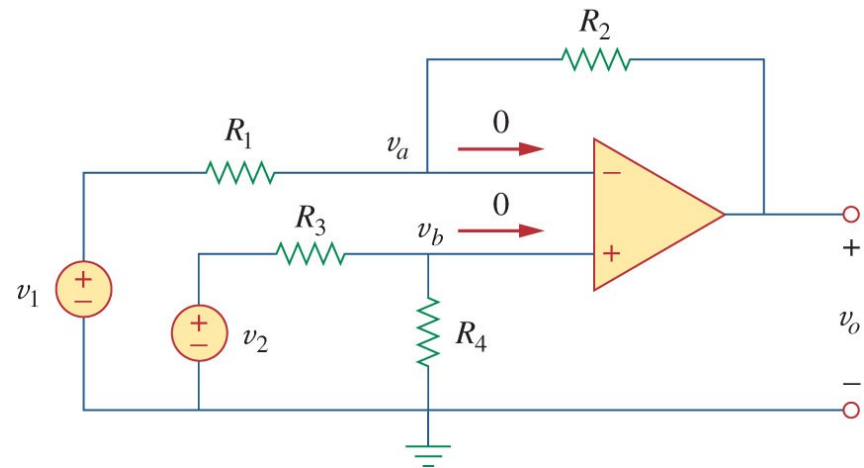
Practice Problems

5.6 Find v_o and i_o



Difference Amplifier

- Subtraction should come naturally to the op-amp since its output is proportional to the difference between the two inputs
- Applying KCL to node a in the circuit shown gives:



$$\frac{v_1 - v_a}{R_1} = \frac{v_a - v_o}{R_2} \Rightarrow v_o = \left(\frac{R_2}{R_1} + 1 \right) v_a - \frac{R_2}{R_1} v_1$$

Difference Amplifier

- Applying KCL to node b gives:

$$\frac{v_2 - v_b}{R_3} = \frac{v_b - 0}{R_4} \Rightarrow v_o = \frac{R_4}{R_3 + R_4} v_2$$

- But $v_a = v_b$ resulting in the following relationship:

$$v_o = \frac{R_2(1 + R_1/R_2)}{R_1(1 + R_3/R_4)} v_2 - \frac{R_2}{R_1} v_1$$

Common Mode Rejection

- It is important that a difference amplifier reject any signal that is common to the two inputs
- For the given circuit, this is true if:

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

- At which point, the output is:

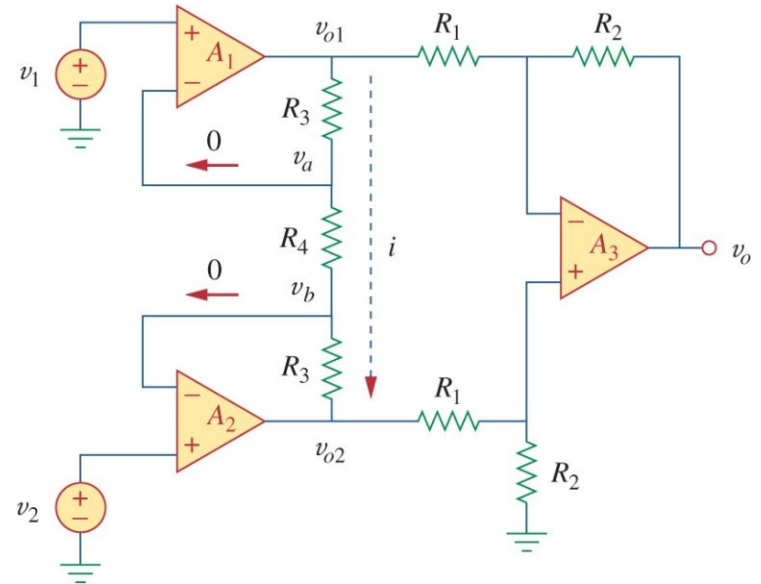
$$v_o = \frac{R_2}{R_1} (v_2 - v_1)$$

Instrumentation Amplifier

The difference amplifier has one significant drawback:

- The input impedance is low

By placing a non-inverting amplifier stage before the difference amplifier this can be resolved

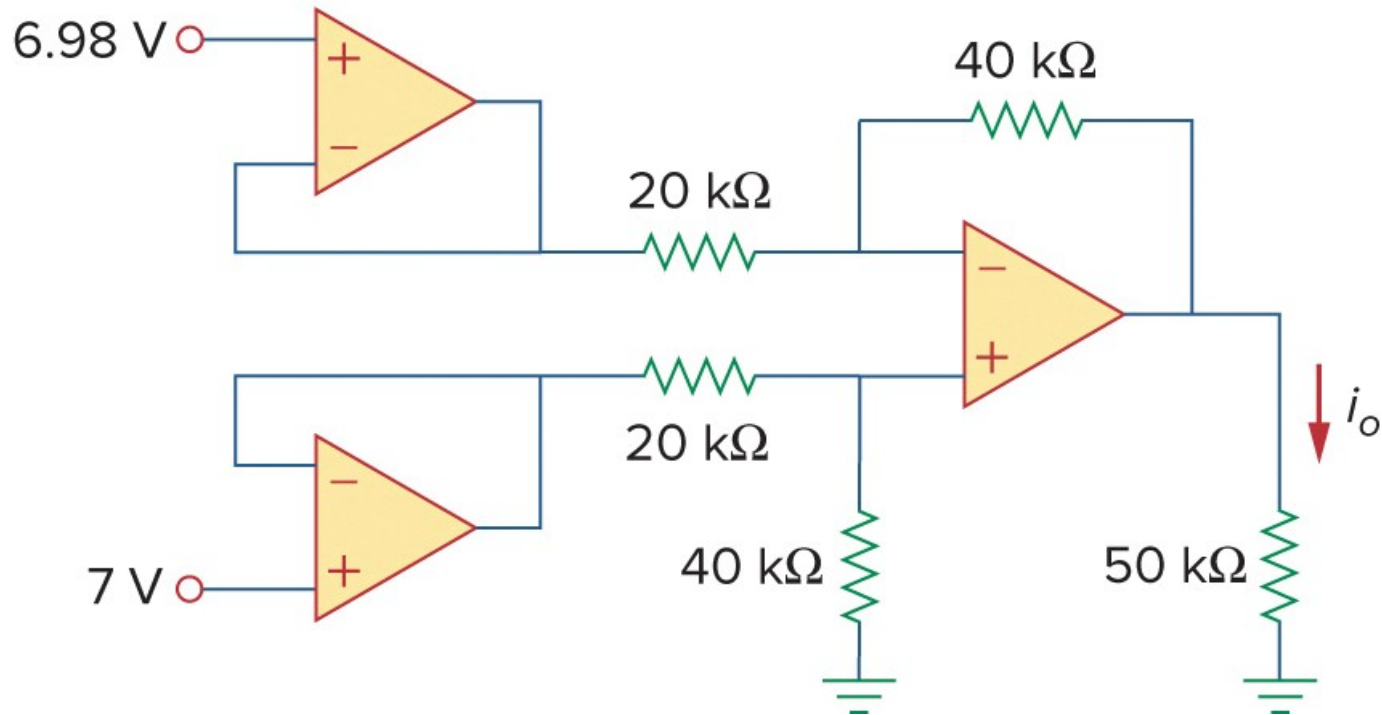


Instrumentation Amplifier

- Instrumentation amplifiers are so useful, they are often packaged as a single chip with the only external component being the gain resistor
- They are very effective at extracting a weak differential signal out of a large common mode signal
- In circuits exposed to external electrical noise, this is important in order to maintain a high signal-to-noise ratio

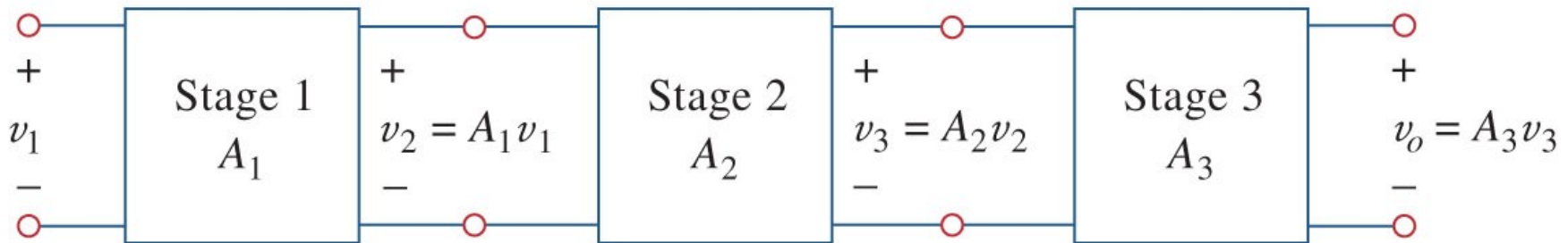
Practice Problems

5.8 Find i_o in this instrumentation amplifier



Cascaded Op Amps

- It is common to use multiple op-amp stages chained together
- This head to tail configuration is called “cascading”
- Each amplifier is then called a “stage”



Cascaded Op Amps

- Due to the ideal op-amps's input and output impedance, stages can be chained together without impact the performance of any one stage
- One reason to cascade amplifier stages is to increase the overall gain
- The gain of a series of amplifiers is the product of the individual gains:

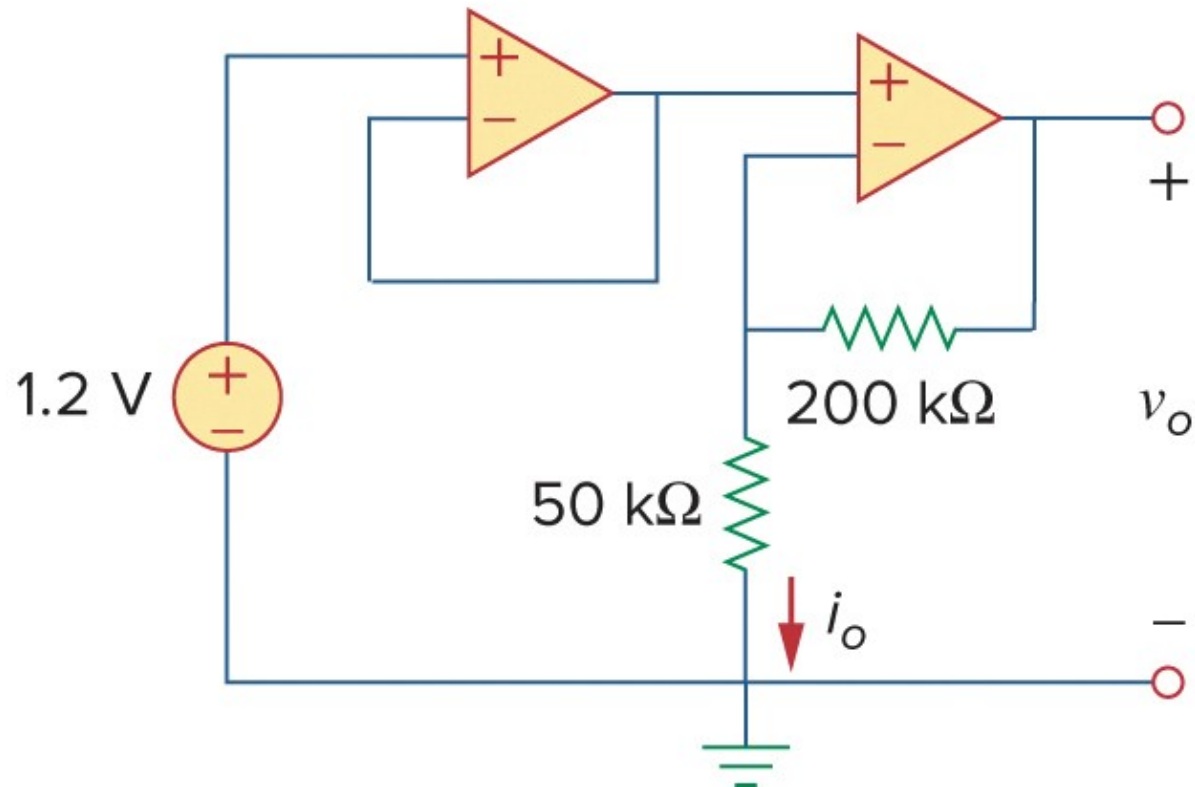
$$A = A_1 A_2 A_3$$

Cascaded Amplifiers

- For example, two stages each having a gain of 100, have a combined gain of 10,000
- Mixing high gain and improved input impedance is another reason to cascade

Practice Problems

5.9 Find v_o and i_o

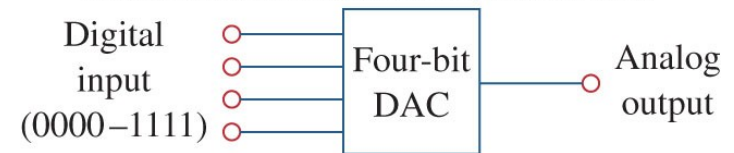


Digital to Analog Converter

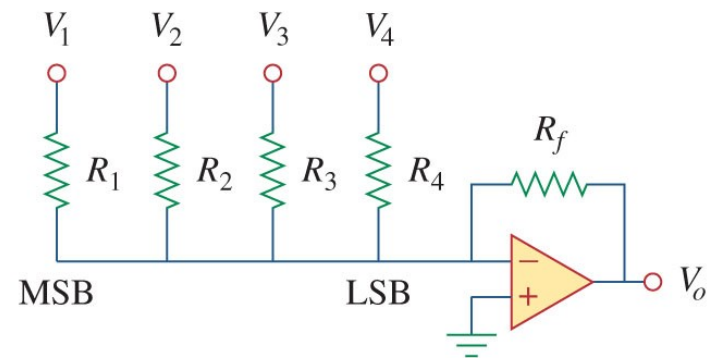
- The summing amplifier can be used to create a simple digital to analog converter (DAC)
 - Recall that each input has its own multiplier resistor
- In a digital signal, the input voltage will be either zero which represents '0' or a non-zero voltage which represents '1'
- The function of a DAC is to take a series of binary values that represent a number and convert it to an analog voltage

Digital to Analog Converter

- By selecting the input resistors such that each input will have a weighting according to the magnitude of their place value
- Each lesser bit will have half the weight of the next higher bit
- The feedback resistor provides an overall scaling, allowing the output to be adjusted according to the desired range



(a)



(b)